COM3549-FINANCIAL TECHNOLOGIES

Term Project Report: Credit Card Default Prediction

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1. Data Description

1.1. Dataset Source and Objective

The dataset used in this study is derived from research examining the default payments of credit card clients in Taiwan. It is hosted in the UCI Machine Learning Repository as the "Default of Credit Card Clients" dataset.

The primary objective of this project is to predict whether a customer will default on their payment in the following month based on demographic features and past payment history. This is a critical classification problem for financial risk management.

1.2. Variables and Dimensions

The dataset consists of **30,000 observations (N)** and **24 variables (k)**, including the target variable. The response variable is binary: default.payment.next.month (Yes = 1, No = 0).

The explanatory variables ($X_1 – X_{23}$) are defined as follows:

- **X1 (LIMIT_BAL):** Amount of the given credit (NT dollar). It includes both the individual consumer credit and his/her family (supplementary) credit.
- **X2 (SEX):** Gender (1 = male; 2 = female).
- **X3 (EDUCATION):** Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
- **X4 (MARRIAGE):** Marital status (1 = married; 2 = single; 3 = others).
- **X5 (AGE):** Age (year).
- **X6 - X11 (PAY_0 ... PAY_6):** History of past payment. Tracked the past monthly payment records (from April to September, 2005).

- Measurement scale: -1 = pay duly; 1 = payment delay for one month; 2 = payment delay for two months; . . .; 9 = payment delay for nine months and above.
- *X6*: Repayment status in September, 2005.
- *X11*: Repayment status in April, 2005.

- **X12 - X17 (BILL_AMT1 ... BILL_AMT6):** Amount of bill statement (NT dollar). From September 2005 (X12) to April 2005 (X17).
- **X18 - X23 (PAY_AMT1 ... PAY_AMT6):** Amount of previous payment (NT dollar). From September 2005 (X18) to April 2005 (X23).

2. Descriptive Statistics

A summary of the numerical and categorical features of the dataset is presented below. Data cleaning procedures were applied prior to analysis to improve data quality.

2.1. Data Cleaning

Upon examining the raw dataset, undocumented labels (such as 0, 5, 6) were detected in the EDUCATION and MARRIAGE columns. To ensure model consistency, rows containing these unknown categories were either removed or merged into the "Other" category (4 or 3). Additionally, **SMOTE** (Synthetic Minority Over-sampling Technique) was applied during the training phase to address class imbalance.

2.2. Statistics for Numerical Variables

Distribution measures for key numerical variables such as Credit Limit (LIMIT_BAL) and Age (AGE) are as follows:

Variable	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
LIMIT_BAL	167,484	140,000	129,747	10,000	1,000,000	0.99 (Right Skewed)	0.54
AGE	35.48	34.00	9.21	21	79	0.73 (Right Skewed)	0.04
BILL_AMT1	51,223	22,381	73,635	-165,580	964,511	2.66 (Highly Skewed)	9.81

Comment: The high standard deviation and positive skewness in LIMIT_BAL and Bill Amounts indicate that there are a small number of clients with very high limits, resulting in a right-skewed distribution.

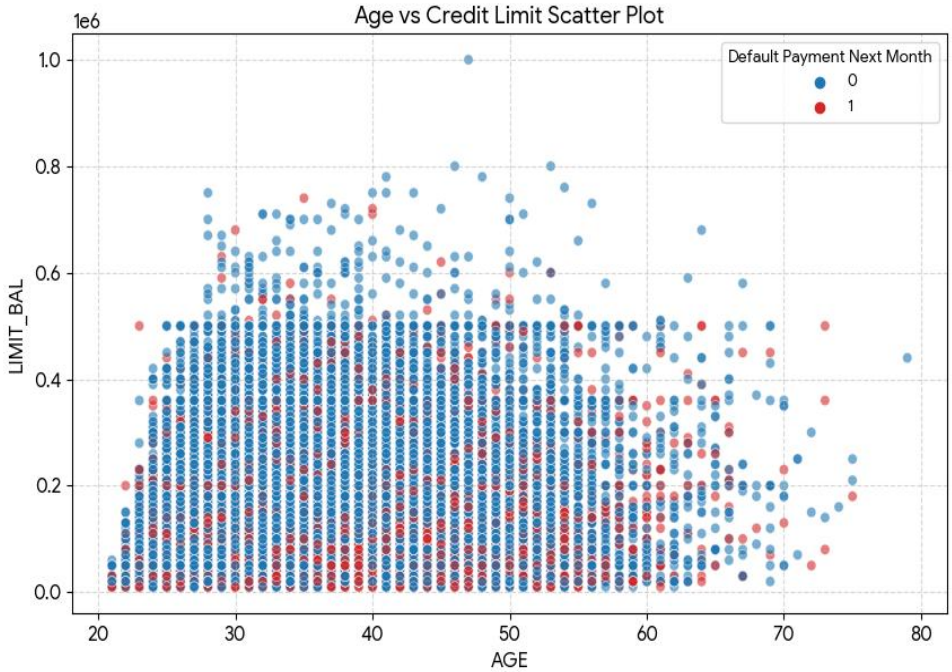
2.3. Categorical Variables (Frequency Tables)

Examining the distribution of the target variable default reveals that the dataset is imbalanced.

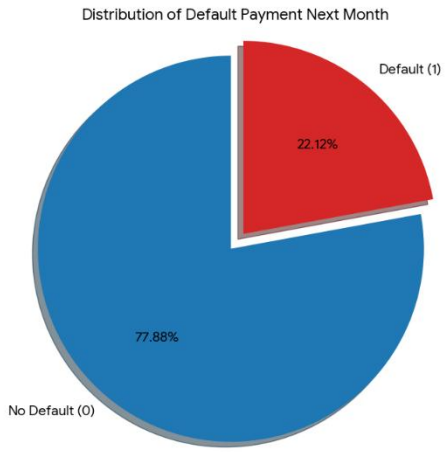
Class (Default)	Frequency	Percentage (%)
0 (No Default)	23,364	77.88%
1 (Default)	6,636	22.12%
Total	30,000	100.00%

3. Graphical Representation

Univariate and multivariate visualizations were created to understand the relationships within the dataset.



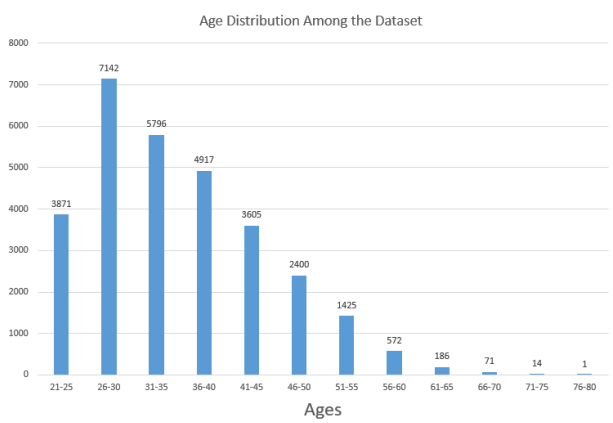
The chart reveals that defaults (red) are concentrated in lower credit limits and younger age groups, while non-defaulters (blue) dominate as age and limits increase.

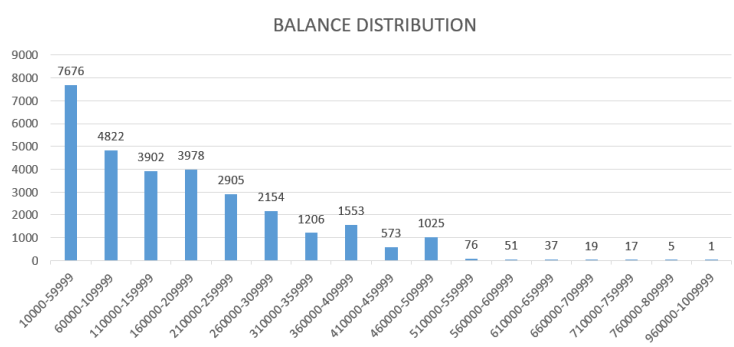


This chart shows that while defaults (red) are concentrated in lower credit limits and younger age groups, the proportion of non-defaulters (blue) increases with higher age and credit limits.

3.1. Histograms (Distribution Analysis)

The histogram for the AGE variable shows that the majority of customers are clustered in the 25-35 age range. The number of customers decreases as age increases. Similarly, the LIMIT_BAL histogram confirms that lower limit cards are in the majority.



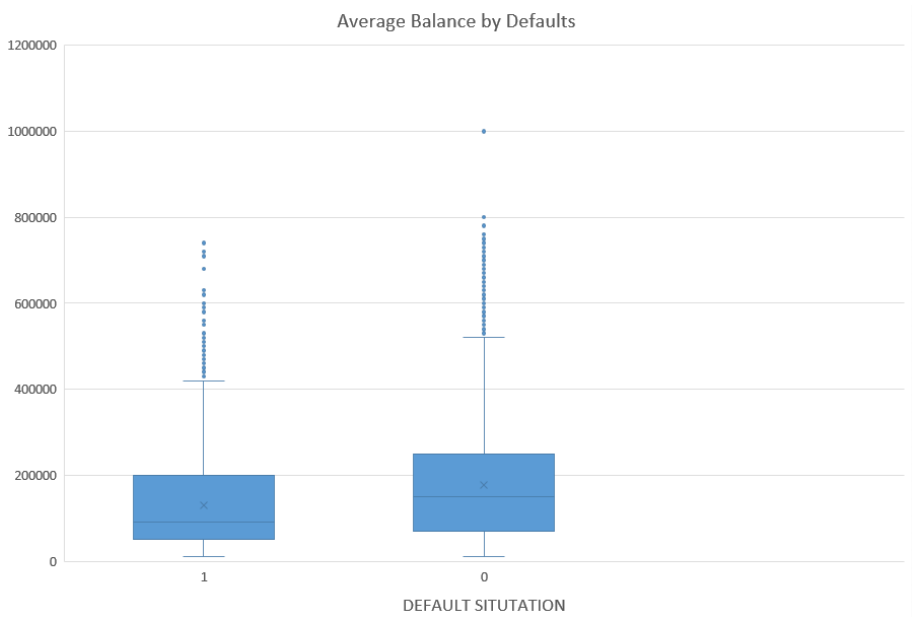


3.2. Boxplot (Outlier Analysis)

A Boxplot analysis of LIMIT_BAL against Default status was conducted:

- The median credit limit for non-defaulters (0) was observed to be higher than that of defaulters (1). This points at the fact that the permitted balance limit is higher in average for non-defaulters compared to defaulters.

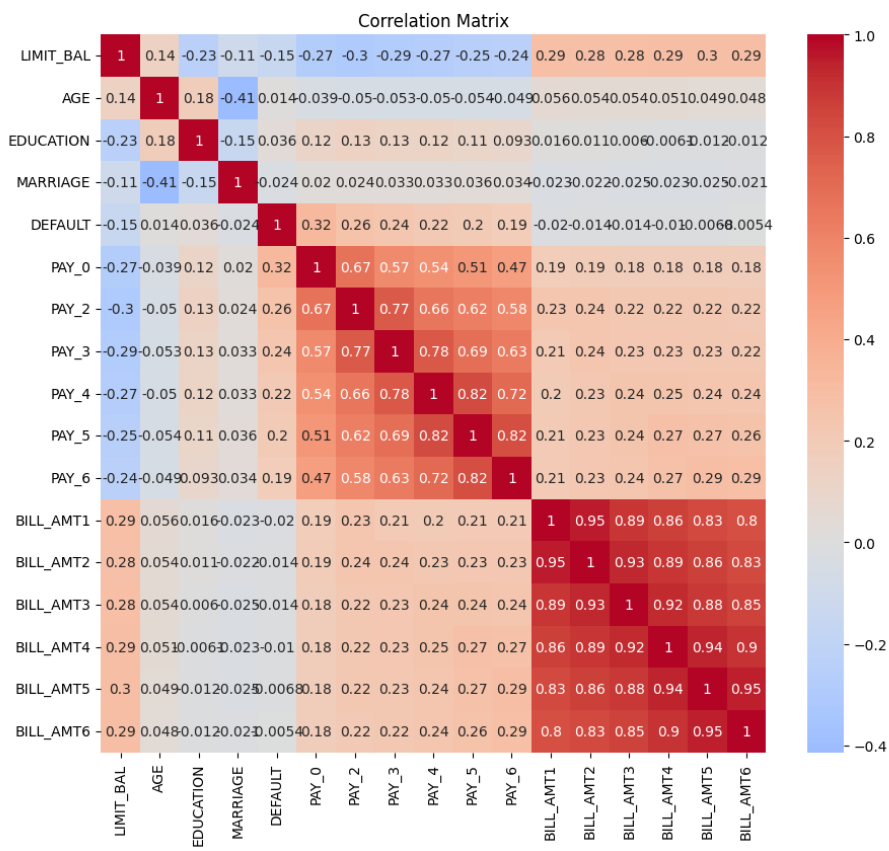
- A significant number of outliers exist above the upper whisker in both groups. These represent the high-net-worth segment and were preserved during the data cleaning phase.



3.3. Correlation Matrix (Heatmap)

When examining relationships between variables:

- A very high positive correlation (over 0.90) exists between BILL_AMT variables. This is expected as debt is cumulative.
- PAY_0 ... PAY_6 (Payment status) variables showed the highest correlation with the target variable (default). This indicates that payment delays are the strongest predictor of default.

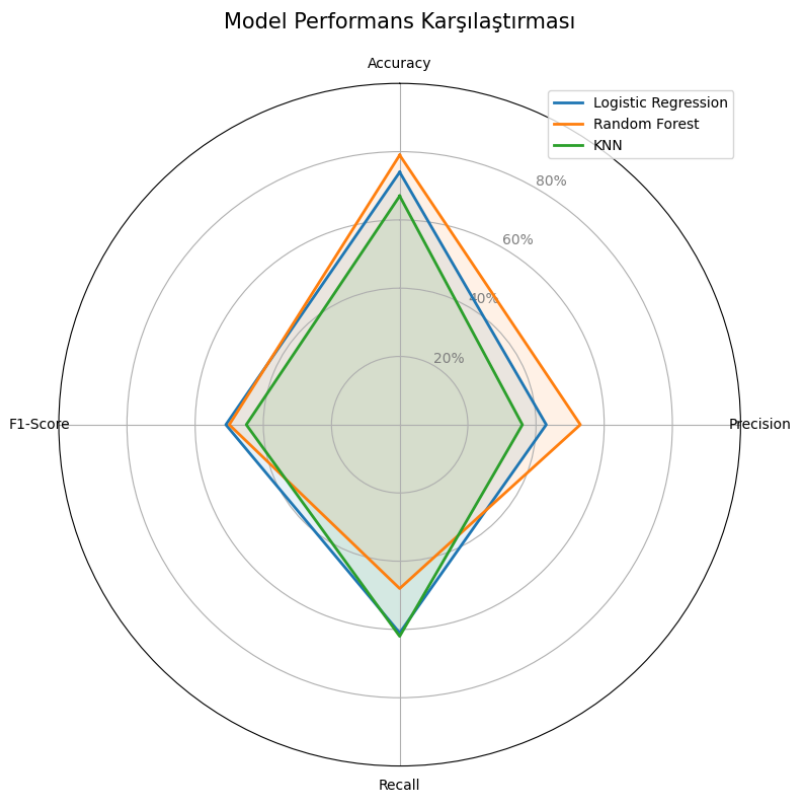


4. Results & Comparison

Three fundamental machine learning algorithms were used in this project: **Logistic Regression**, **Random Forest**, and **K-Nearest Neighbors (KNN)**. Model performance was evaluated using Accuracy, Precision, Recall, and F1-Score metrics.

4.1. Model Performance Table

Model	Accuracy	Precision (1)	Recall (1)	F1-Score (1)
Logistic Regression	0.76	0.48	**0.61**	**0.51**
Random Forest	**0.79**	0.51	0.48	0.49
KNN (k=5)	0.67	0.36	0.62	0.45



4.2. Comparison and Interpretation

- **Accuracy:** Random Forest provided the highest overall accuracy at 79%. However, since the dataset is imbalanced (the majority class is "non-default"), relying solely on accuracy can be misleading.
- **Risk Management Perspective:** For financial institutions, the primary risk is failing to identify a customer who will not pay (False Negative). Therefore, the **Recall** metric is critical.
- **Logistic Regression:** With a Recall of 61% and a balanced F1 score, this was the most successful model in capturing risky customers.

4.3. Conclusion

The analysis concludes that while Random Forest is more successful in general classification (Accuracy), **Logistic Regression** is more stable in its capacity to capture potential bad debts (defaults) when distinguishing between "Credible or Not Credible" clients. In future studies, more complex models such as Artificial Neural Networks (ANN)—as mentioned in the original literature (Yeh & Lien, 2009)—or XGBoost could be tested to improve success rates.

5. References

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